



EDUCATION

Bachelor of Computer Engineering, Major in Game Technology

Electronic Engineering Polytechnic Institute of Surabaya (2020 - 2024)
GPA 3.67 (Cumlaude)

TECHNICAL SKILLS

- Python (NumPy, scikit-learn, PyTorch)
- Reinforcement Learning (Q-Learning, FSM)
- Fuzzy Logic & Recommender Systems
- Data Scraping, Analysis, Visualization (Selenium, Gephi, Tableau)
- C++ | C# (engine-side AI integration)
- Applied research & evaluation studies

CONTACT

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Putih, Jakarta Pusat, DKI Jakarta
10510

KENAN FIRMANSYAH

AI / ML ENGINEER

Applied AI/ML engineer with five peer-reviewed publications on adaptive game AI, dynamic difficulty, and context-aware recommender systems. Industry experience as a data scientist plus production AI work in shipped interactive titles.

SELECTED PUBLICATIONS

Adaptive Radio System Using MetaSounds for Serious VR Game
IEEE Access (Vol. 13), Nov 2, 2025 | DOI: 10.1109/ACCESS.2025.3627544

- FSM-driven adaptive audio across affective states; n=30 VR user study, rated more immersive than non-adaptive baseline.

Dynamic Level of Difficulties Using Q-Learning and Fuzzy Logic
IEEE Access (Vol. 12), Sep 12, 2024 | DOI: 10.1109/ACCESS.2024.3457801

- Hybrid Q-learning + fuzzy DDA system; **+35% session length, +28% player effectiveness.**

Context-Aware Recommender System for In-Game Radio
INASS IJIES, Sep 2, 2024 | DOI: 10.22266/ijies2024.1031.26

- Deep-learning hybrid filtering on contextual signals; **F1 up to 0.71**, outperforming baselines.

EXPERIENCE

Game Developer (AI/Gameplay)
Agate | 2026 Feb - 2026 May

- Production game-AI work at one of Indonesia's largest studios on an unannounced title.

Data Scientist (Intern)
PT. eBdesk | 2023, 6 months

- Built scraping, analysis, and visualization pipelines in Python (Selenium, Gephi, Tableau).

ADDITIONAL RESEARCH & APPLIED AI WORK

1 APPLIED AI IN SHIPPED & RESEARCH PROJECTS

Xanthous — AI in VR Horror (Thesis)

EEPIS | 2023 Feb - 2024 July | Unreal Engine

- Designed and implemented enemy AI systems and gameplay mechanics for a serious VR horror game; underlying research published.

Until Dawn Do Us Part — Enemy AI & System Programmer

iPlay Apple Developer Academy Game Jam 2025 | 1st Place

- Wave-based enemy AI and game systems for a survival twin-stick shooter, shipped in 2 days.

AI Programmer roles in Game Jams

Brackeys 2024.2 (Unreal AI) | Garena GameJam (Unreal AI Perception, stealth)

- Specialized AI/perception work across jam-scale prototypes.

2 ADDITIONAL PUBLICATIONS

Real Time Procedural Audio for Immersive Gameplay with MetaSounds

TCCE 2025, Trends in Computational and Cognitive Engineering

Dynamic Day and Night Cycle Impact in a Serious VR Game

ICVR 2024, International Conference on Virtual Reality | +25% immersion improvement reported.

3 OTHER ENGINEERING EXPERIENCE

Apple Developer Academy @ BINUS (2025 Feb - 2025 Dec) — Interdisciplinary product development; UI/UX, prototyping, iterative testing.

Lead Developer, Starpixel Studios (2024 Oct - 2025 April) — Multiplayer horror game; Blueprint + C++ for performance-critical systems.

JoyBait Studio (Saturated) (2025 Oct - 2025 Dec) — Project manager and programmer on a first-person horror game shipped to Mac App Store.

Home Server (Personal, 2023) — Proxmox-based multi-purpose server (NAS, database, compute).

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